

Notes on Continuum Mechanics (an excerpt)

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1.5 Algebraic Operations with Tensors

1.5.1 Dyadic

The *tensor product*, made up of two vectors $\vec{v}$ and $\vec{u}$, becomes a dyad, which is a particular case of a second-order tensor. The dyad is represented by:

$$\vec{u}\vec{v} \equiv \vec{u} \otimes \vec{v} = \mathbf{A} \quad (1.72)$$

where the operator $\otimes$ denotes the *tensor product*. Then, we define a *dyadic* as a linear combination of dyads. Furthermore, as we will see later, any tensor can be represented by means of a linear combination of dyads, (see Holzapfel (2000)). The tensor product has the following properties:

1.

$$(\vec{u} \otimes \vec{v}) \cdot \vec{x} = \vec{u}(\vec{v} \cdot \vec{x}) \equiv \vec{u} \otimes (\vec{v} \cdot \vec{x}) \quad (1.73)$$

2.

$$\vec{u} \otimes (\alpha\vec{v} + \beta\vec{w}) = \alpha\vec{u} \otimes \vec{v} + \beta\vec{u} \otimes \vec{w} \quad (1.74)$$

3.

$$\begin{aligned} (\alpha\vec{v} \otimes \vec{u} + \beta\vec{w} \otimes \vec{r}) \cdot \vec{x} &= \alpha(\vec{v} \otimes \vec{u}) \cdot \vec{x} + \beta(\vec{w} \otimes \vec{r}) \cdot \vec{x} \\ &= \alpha[\vec{v} \otimes (\vec{u} \cdot \vec{x})] + \beta[\vec{w} \otimes (\vec{r} \cdot \vec{x})] \end{aligned} \quad (1.75)$$

where α and β are scalars. By definition, the dyad does not contain the commutative property, i.e., $\vec{u} \otimes \vec{v} \neq \vec{v} \otimes \vec{u}$.

The equation (1.72) can also be expressed in the Cartesian system as:

$$\begin{aligned} \mathbf{A} = \vec{u} \otimes \vec{v} &= (\mathbf{u}_i \hat{\mathbf{e}}_i) \otimes (\mathbf{v}_j \hat{\mathbf{e}}_j) \\ &= u_i v_j (\hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j) \quad (i, j = 1, 2, 3) \quad \#(1.76) \end{aligned}$$

$$\begin{aligned} &= A_{ij} (\hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j) \\ \underbrace{\mathbf{A}}_{\text{Tensor}} &= \underbrace{A_{ij}}_{\text{components}} \underbrace{\hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j}_{\text{basis}} \quad (i, j = 1, 2, 3) \quad \#(1.77) \end{aligned}$$

In this textbook, the components of a second-order tensor can be represented in different ways, namely:

$$\begin{array}{c}
 \underbrace{A = \vec{u} \otimes \vec{v}} \\
 \downarrow \\
 \text{components} \\
 \downarrow \\
 (\mathbf{A})_{ij} = (\vec{u} \otimes \vec{v})_{ij} = u_i v_j = A_{ij}
 \end{array} \tag{1.78}$$

These components are explicitly expressed in matrix form as: